Yasaswi Bhumireddy

AI ENGINEER | GENERATIVE AI & RELIABLE ML SYSTEMS

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PROFESSIONAL SUMMARY

AI engineer with experience delivering generative AI platforms at AWS and machine learning systems in financial services. Builds retrieval and agent workflows, asynchronous Python services, evaluation pipelines, and observable cloud deployments. Focuses on measurable model behavior, reliable execution, and responsible AI delivery.

TECHNICAL EXPERTISE

Languages & services: Python, SQL, TypeScript, FastAPI, AsyncIO, REST APIs
AI & ML: RAG, LLM evaluation, agent workflows, tool calling, guardrails, PyTorch, Hugging Face, LangChain, LlamaIndex, XGBoost, LightGBM
Cloud & data: AWS, Amazon Bedrock, AgentCore, SageMaker, S3, PostgreSQL, Redis, OpenSearch, PySpark, Kafka, Flink
Delivery & operations: Docker, Kubernetes, CI/CD, AWS CDK, MLflow, Airflow, OpenTelemetry, CloudWatch, model monitoring

PROFESSIONAL EXPERIENCE

Delivery Consultant - Gen AI/ML | Amazon Web Services

February 2025 - Present

- Deliver enterprise generative AI solutions spanning Amazon Bedrock, AgentCore, retrieval, tool calling, model evaluation, and cloud deployment; collaborate with client engineering and business stakeholders on architecture and release decisions.
- Architected a self-service platform for an e-commerce and retail client handling 2.1M monthly conversations and reducing repeat contacts by 18%, with human escalation for high-risk cases.
- Build asynchronous Python services with bounded pools, deadlines, retries, and cancellation to support model and tool adapters under burst traffic; contributed to 99.9% availability targets.
- Benchmark conversational and tool-using workflows for accuracy, safety, latency, and cost. Applied prompt caching, batching, and context compaction to reduce serving cost by 28%.
- Fine-tuned Amazon Nova models on 68K de-identified client cases, improving structured-action accuracy from 84% to 92%; added experiment tracking and validation gates before release.
- Implement typed tool interfaces, approval gates, durable workflows, canary deployment, and rollback. Connect OpenTelemetry, CloudWatch, and X-Ray signals to model and customer-outcome metrics.

Machine Learning Engineer | Jio Payments Bank

January 2020 - July 2022

- Built fraud and dispute analytics workflows using gradient-boosted models, graph features, and streaming rules; improved confirmed-fraud recall by 14% and reduced false-positive reviews by 19%.
- Engineered point-in-time feature pipelines with Kafka, Flink, PySpark, PostgreSQL, and Redis, including schema lineage, replay, and late-event controls.
- Built low-latency Python scoring APIs and standardized model versioning, deployment, drift monitoring, and rollback with MLflow, Airflow, Docker, and Kubernetes.

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AI Engineering | Selected work and education

INDEPENDENT ENGINEERING PROJECTS

Runnable portfolio implementations with tests and documented limitations. Demonstration datasets are synthetic or hand-authored; project metrics are separate from employment outcomes.

Evidence Search | Python, FastAPI, SQLite

- Built a tenant-scoped retrieval API with reader/writer authorization, overlapping document chunks, BM25 ranking, content-versioned citations, and persistent document storage.
- Added integration tests covering invalid credentials, cross-tenant reads and deletes, spoofed tenant inputs, persistence, document versions, and content-free audit records.

LLM Evaluation Gates | Python, Evaluation, Release Governance

- Implemented paired evaluation with dataset fingerprints, slice-level accuracy, bootstrap intervals, citation rules, p95 latency, and request-cost budgets.
- Created an 18-case synthetic benchmark demonstrating how a refusal regression can block a candidate despite improved aggregate accuracy.

Durable Agent Workflow | Python, SQLite, State Machines

- Implemented persistent approval-gated execution, idempotency-key validation, atomic local state transitions, and audit events; tested replay, restart recovery, and rollback.

Resilient Inference Service | Python, AsyncIO

- Built a provider adapter with bounded concurrency, end-to-end deadlines, transient retries, cancellation propagation, and a process-local circuit breaker.
- Added a reproducible synthetic workload benchmark reporting queue-inclusive p50/p95 latency, successful request throughput, failures, and peak concurrency.

Additional projects: Fraud Risk Lab; Demand Forecast Backtester; Drift Sentinel; Catalog Recommender; Ticket Triage Lab; Privacy Redaction Gateway.

EDUCATION

Master of Science in Data Science | University of North Carolina

August 2023 - December 2024

SELECTED RESEARCH

Genomic Segmentation with Multimodal and Interdimensional Advances

AI-Powered Chatbots for Democratizing Manufacturing Education

CERTIFICATIONS AND PROFESSIONAL LEARNING

Meta Back-End Developer and Front-End Developer Professional Certificates - Coursera
IBM Full Stack Software Developer; DevOps and Software Engineering Professional Certificates - Coursera
Build REST APIs with FastAPI; Become a RESTful API Developer - LinkedIn Learning